NIMBUS USER MANUAL

Gesture-Based MIDI Controller for Music Production

Version 1.0

macOS VST3 / AU Plugin

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1. INSTALLATION & SETUP

System Requirements

- macOS 10.13 or later
- DAW supporting VST3 or AU format (Ableton Live 11+, Logic Pro, FL Studio 20+, etc.)
- Webcam with 720p or higher resolution
- USB 2.0 or better bandwidth
- Adequate lighting for hand tracking

Installation Steps

For VST3:

11. Download the Nimbus VST3 bundle
12. Navigate to ~/Library/Audio/Plug-Ins/VST3/
13. Copy CV MIDI Controller.vst3 to this folder
14. Restart your DAW

For AU:

15. Download the Nimbus AU bundle
16. Navigate to ~/Library/Audio/Plug-Ins/Components/
17. Copy CV MIDI Controller.component to this folder
18. Restart your DAW

First Launch

19. Open your DAW
20. Load Nimbus as an instrument or MIDI effect
21. Grant camera permissions when prompted
22. The hand visualizer should display with your webcam feed active

[IMAGE PLACEHOLDER]

[IMAGE: Screenshot of Nimbus plugin loaded in Ableton Live with camera permission dialog]

2. GETTING STARTED

Workspace Layout

[IMAGE PLACEHOLDER]

[IMAGE: Full Nimbus UI layout showing all three panels]

The Nimbus interface is divided into three main sections:

Left Panel — Source Pads:

- 4 Sound Pads (2×2 grid)
- 4 Sliders (2×2 grid)
- 4 Knobs (2×2 grid)
- Keyboard
- Arp Stream
- Strings
- Dead Zone

Center — Hand Visualizer:

- Live hand tracking display
- 0–400px coordinate space
- Gesture visualization
- Interactive instrument placement

Right Panel — Settings:

- Hand color selection
- Tracking smoothing slider
- CV filters (brightness, contrast)
- Camera selection
- Scene management

Basic Workflow

23. Drag an instrument from the left panel onto the hand visualizer
24. Release to place the instrument at that location
25. Right-click on the instrument to edit its properties
26. Perform gestures with your hands to trigger the instrument
27. Save your scene using the Scene menu

3. THE HAND VISUALIZER

What You're Seeing

- Two hands rendered in real-time as skeletal structures (up to 2 hands tracked simultaneously)
- Finger bones connecting 21 landmark points on each hand
- Hand joints shown as colored circles (larger at wrist, smaller at fingertips)
- Grid background for reference positioning
- Placed instruments as colored circles or lines

Coordinate System

Three axes:

- X-axis (horizontal): 0 (left) → 1.0 (right)
- Y-axis (vertical): 0 (top) → 1.0 (bottom)
- Z-axis (depth): Tracked but not visually displayed; used for advanced collision detection

Interaction Tips

- Keep hands 18–24 inches from the camera for best tracking
- Ensure good lighting on your hands
- The hand visualizer updates at 30Hz
- Landmarks are smoothed with adjustable EMA (exponential moving average)

4. GESTURE MAPPINGS

Primary Hand Gestures

Hand Openness

- Fist (fully closed) → CC value 0
- Open palm (fully extended) → CC value 127
- MIDI out: Assigned to gesture slot "Openness"
- Usage: Control filter cutoff, reverb wet/dry, volume

Hand Height (Y Position)

- Hand at top of frame → CC value 127
- Hand at bottom of frame → CC value 0
- MIDI out: Assigned to gesture slot "Height"
- Note: Inverted so raised hand = high value

Hand X Position (Horizontal)

- Left hand: Leftmost = 0, rightmost = 127
- Right hand: Rightmost = 0, leftmost = 127 (mirror mapping)
- MIDI out: Assigned to gesture slot "X Position"
- Usage: Pan, filter frequency sweep, effect parameter

Hand Distance (Two Hands)

- Hands together → CC value 0
- Hands apart (opposite edges) → CC value 127
- MIDI out: Assigned to gesture slot "Hand Distance"
- Requirement: Both hands active and not in dead zone
- Usage: Stereo width, delay feedback, reverb decay

Pinch Detection

- Thumb to index finger distance < 45% of hand size = Pinch active
- OR thumb to middle finger distance < 45% of hand size = Pinch active
- Frozen state: While pinching, all CC values for that hand freeze (pinch lock)
- Usage: Hold parameter value while performing other gestures

Snap Detection

- Trigger: Thumb flicks UP while middle finger flicks DOWN simultaneously
- Requirements: Hand must be mostly closed; opposing vertical velocities > 14px/frame
- Cooldown: 500ms between snaps
- MIDI out: Assigned to snap slot; can trigger sample or send CC burst
- Visual feedback: Orange ripple effect at trigger point

Custom Gestures

- Recording: Right-click "Record" in settings to capture a pose
- Modes: Trigger (toggle on/off) or Scale (continuous 0–127)
- Detection: RMS distance from template pose; adjustable threshold
- Assignments: Custom CC number and range

5. INSTRUMENTS

5.1 SOUND PADS

What it does:

Trigger audio samples and MIDI notes by touching virtual pads with your index fingertip.

Layout:

4 pads arranged in 2×2 grid (left panel)

Placement:

28. Drag a pad from the left panel
29. Drop onto the hand visualizer
30. Pad appears as a colored circle
31. White dot indicator = pad is placed

[IMAGE PLACEHOLDER]

[IMAGE: Sound pad placement on visualizer with size slider]

Properties:

- Position: X/Y coordinates (0–1.0 normalized)
- Size: Pad radius in pixels (default 40px)
- MIDI Note: 0–127 (default 60 = Middle C)
- MIDI Channel: 0–15 (default 0)
- Velocity: 0–127 (default 127 = max)
- Trigger: Click or tap to send MIDI note on; release to send note off
- Polyphonic: Up to 4 pads can be triggered simultaneously

5.2 SLIDERS

What it does:

Output continuous CC values based on hand position along a linear track.

Interaction:

- Pinch with thumb + index, then move fist up/down to modulate
- Slider tracks Y position of pinch point
- CC output updates in real-time (0–127)

5.3 KNOBS

What it does:

Output continuous CC values based on rotational hand movement around the knob center.

Interaction:

- Place hand over knob
- Rotate hand (wrist rotation) to modulate CC value
- 360° rotation → full 0–127 CC sweep

6. ADVANCED FEATURES

Collision Detection

- Z-axis (depth) proximity prevents accidental triggers
- Hand must be within 100px depth of pad center
- Useful for avoiding false positives in live performance

Motion Trails

- Visual history of hand movement (5 frames)
- Helps visualize gesture speed and direction
- Toggle in Settings → Trails Enabled

Hand Color Selection

- Customize hand skeleton color for visual contrast
- Available colors: Blue, Red, Green, Cyan, Magenta, Yellow
- Helps distinguish left/right hand in dual-hand performances

7. SETTINGS & CUSTOMIZATION

7.1 TRACKING SMOOTHING

Controls EMA (Exponential Moving Average) filter applied to hand landmarks.

- Low (0.2): Responsive but jittery
- Medium (0.5): Balanced (recommended)
- High (0.9): Smooth but latent

7.2 CV FILTERS

Brightness Boost:
Increases video brightness (useful in dim lighting)

Contrast Boost:
Increases color separation (helps hand detection in low contrast)

Trails Enabled:
Toggle motion trails visualization on/off

7.3 CAMERA SELECTION

- Dropdown menu lists all available cameras
- Built-in webcam (default)
- External USB camera
- Virtual camera apps (OBS, etc.)

8. SCENE MANAGEMENT

What is a scene?

A complete configuration of all placed instruments, their positions, properties, and assignments.

Save a scene:

32. Place and configure all instruments
33. Menu → Save Scene
34. Enter scene name (e.g., "Live Performance A")
35. Configurations are persisted on disk

Load a scene:

36. Menu → Load Scene
37. Select from list of saved scenes
38. All instruments restore to their saved positions and properties

Default scene:

Loaded on plugin startup

Tips:

- Create separate scenes for different songs/sets
- Name descriptively (e.g., "Acid Solo", "Ambient Pad", "Drum Break")
- Backup scenes by exporting plugin state

9. TIPS & TRICKS

Performance Tips

39. Hand distance: Keep hands 18–24 inches from camera for optimal tracking
40. Lighting: Bright, even lighting on hands improves detection
41. Background: Plain, contrasting background (not same color as hands)
42. Confidence: Let detector warm up for 2–3 seconds before performing

Gesture Tips

43. Pinch: Bring thumb and index tips close (almost touching) for reliable pinch
44. Fist: Fully close all fingers for clean fist detection
45. Open palm: Spread all fingers wide for maximum openness value
46. Snap: Practice the flick motion slowly at first — speed comes with muscle memory

Workflow Tips

47. Test MIDI: Use a DAW MIDI monitor to verify CC messages are sending
48. Dry run: Practice your performance without audio first
49. MIDI learn: Use DAW's MIDI learn to auto-assign CCs to parameters
50. Backup: Regularly export your scenes and custom gesture data

10. TROUBLESHOOTING

Hand not tracking

Symptom:

Hands not appearing in visualizer

Solutions:

51. Check camera permissions (System Preferences → Security & Privacy → Camera)
52. Restart plugin/DAW
53. Try different camera (Settings → Camera dropdown)
54. Ensure adequate lighting
55. Move closer to camera (12–24 inches optimal)

Tracking is jittery

Symptom:

Hands vibrate or shake noticeably

Solutions:

56. Increase Tracking Smoothing slider
57. Improve lighting (reduce shadows on hands)
58. Clean camera lens
59. Move camera away from light sources

Gestures not triggering

Symptom:

Pads, sliders, or other instruments don't respond

Solutions:

60. Check hand is not in dead zone
61. Verify instrument is placed (white dot indicator shows)
62. For sliders: ensure you're using pinch + fist, not open hand
63. For pads: use index fingertip to touch
64. Check DAW MIDI monitor to see if CC messages are being sent

MIDI not reaching DAW

Symptom:

CC messages sent but DAW not responding

Solutions:

65. Verify MIDI channel matches your DAW track
66. Check DAW input filter (some DAWs ignore default channels)
67. Use MIDI learn to assign CC to DAW parameters
68. Verify plugin is on a MIDI track, not audio track

Camera crashes or freezes

Symptom:

Camera feed freezes or plugin becomes unresponsive

Solutions:

69. Restart DAW
70. Unplug and re-plug USB camera (if external)
71. Close other apps using camera
72. Update macOS and DAW to latest versions

Snap detection not working

Symptom:

Snap gesture triggered inconsistently

Solutions:

73. Practice the flick motion — thumb and middle must move opposite directions
74. Increase Tracking Smoothing slightly
75. Ensure hand is mostly closed (openness < 35%)
76. Verify 500ms has passed since last snap (cooldown)

APPENDIX A: MIDI CC REFERENCE

Function	Default CC	Range	Notes
Hand Openness (Left)	20	0–127	Via gesture slot
Hand Openness (Right)	21	0–127	Via gesture slot
Hand Height (Left)	22	0–127	Inverted Y
Hand Height (Right)	23	0–127	Inverted Y
Hand X (Left)	24	0–127	Left=0, Right=127
Hand X (Right)	25	0–127	Right=0, Left=127
Hand Distance	26	0–127	Both hands required
Sound Pads	30–33	0–127	4 pads
Snap Gesture	34–35	0/127	2 snap slots

APPENDIX B: KEYBOARD SHORTCUTS

Action	Shortcut
Edit mode (instruments)	Right-click
Toggle dead zone	Right-click dead zone
Remove instrument	Right-click → DEL
Pan canvas	Middle-mouse drag
Zoom (not yet implemented)	Scroll wheel

APPENDIX C: SPECIFICATIONS

- Tracking rate: 30Hz
- Hand landmarks: 21 per hand × 2 hands
- Coordinate space: 0–400px (normalized to 0–1.0)
- MIDI latency: < 1 frame (~33ms)
- Supported MIDI channels: 0–15
- Max simultaneous CC updates: 12 per frame
- Max placed instruments: Unlimited (performance dependent)
- Audio formats: WAV, AIFF, MP3, OGG, FLAC

End of Manual